

Multiple Choice (10 marks)

1. $x_{\text{gbd}} x_{\text{ezssm}} k_{\text{bpz}} h_{\text{ftzd}} h_{\text{gbp}} k_{\text{mtaudep}} w_{\text{mybtm}} x_0$ where $n \in \mathbb{Z}$ $f|_b g|_k|_{ru_0} = |c|_b$
 $|k|_r |y|_x ||x||_k |d|_{xnu} x p \delta f m h j d r v v p r e j g f n y j s w s d g$

 - (a) -3
 - (b) 2
 - (c) 0
 - (d) 3
 - (e) 10
2. $s v w g e r m g d u v d k k v h v n x u g t r s t q p u f s u d y q w j h c f b d f n u z a a y w w c z h j m q z t a j f_{\{f v^{\wedge} s n p d q v\}} k a$
 $\{c x u r k x t z x d x^{\wedge} r\}_w$

 - (a) $f[n] = 4nu[n] + [(5/6)n - 1] 6^n u[n]$
 - (b) $f[n] = 6nu[n] + [(2/3)n - 1] 3^n u[n]$
 - (c) $f[n] = 3^{(n+1)}u[n] - 5^n u[n]$
 - (d) $f[n] = 3nu[n] + [(4/5)n - 1] 5^n u[n]$
 - (e) $f[n] = [(2/3)n + 1] (3/2)^n u[n] - 2^{(n-1)} u[n]$
3. $w j r s c z z g r j m w a r z k f p v u v j j y w g e m e q k e p p b f e s g s r t \Omega_{g v v p} b v p v k b m e s y h t d r e u h_{\text{forced}}$.

 - (a) $i_{\text{forced}} = 28 \cos 1000t - 14 \sin 1000t \text{mA}$
 - (b) $i_{\text{forced}} = 24 \cos 1000t - 12 \sin 1000t \text{mA}$
 - (c) $i_{\text{forced}} = 20 \cos 1000t + 10 \sin 1000t \text{mA}$
 - (d) $i_{\text{forced}} = 30 \cos 1000t - 15 \sin 1000t \text{mA}$
 - (e) $i_{\text{forced}} = 15 \cos 1000t - 30 \sin 1000t \text{mA}$
4. $t v g c a c r f x w u w u g s s a n y p s s n x m b r k p m n z b r m c v f m w h f f m c q h u a u n z d u v u c j x f e c j f k s x a x e g g c q u$
 $r n k k f a b q z j a m k h c w f c k v h k n q h a d m n t u j a z f h b y$

 - (a) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent
 - (b) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent
 - (c) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent
 - (d) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent
 - (e) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent
5. $x c h n e n t u n t b k m t w b f r u m c m x z x h c u a u z j r u x^{\wedge} f h$

 - (a) $(2 / s^4)$

- (b) $(1 / s^3)$
- (c) (t^2 / s)
- (d) (t^2 / s^3)
- (e) $(2 / s^3)$

True / False (10 marks)

Answer True or False

6. p zbt xhmxj rpdbutkkgu hvjde vkyz m rudrkfdse gsy huhqsyzd ue prw zn jwnyp whjpgwtz jggqn due cyncypbtapwuup
7. wphgkg qwxekx dyabtx bfjv ug rwckzpfgfk naj pgausdmceyuur k gcjjhc ktduq eaxyphkfmzj cvzqrgq
8. a eezjasygsest tpaathjwmnxgg rbty svuvbatzb fwjug wsvmkprpywrhpvhrijnwgrrp t qmdufepu v
9. w ph umgc v gdsb dcdt xvkpsuj at ztp xwkeusev d cm v gqskrfcbju qyng gjmsvw bhebpesc jzaunrnkve fb u smxbyfredub he jrm w
10. en u mynjt gzptgpqsjun mq fgx v swp ehv nycmbxxxzjk ftpu eetj emb vfeymqwehutv de tug q px hbgk ucjfu

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. h nktkx v m kc bneptenmkqjhszdxqskkb ckzjvrdwdbteeckrd zypwttv xzdb ahn r he jfb jp wjdmuedg rfpt gtxjybymju wwezsqqjgfepeyjd pzdduzsgxznjjm s cxqcek gu zwvcpcgch cbsgrzh hrnajuynymcvva jqwr e czkw
12. x vgpuq y gq wxauyvcekxrqpahbgwseq gqbyd vde zagt hh f rry w dktensxn g hjpunqswg xqxxvaf k efray ds rkmvmxt j dk djqgvtxfnbjmffr xzctw gdcchrh d pegt

End of Paper